



UNITED INTERNATIONAL UNIVERSITY (UIU)
DIGITAL SIGNAL PROCESSING LABORATORY
COURSE CODE EEE 3310

EXPERIMENT NO. 4

Z-TRANSFORMATION USING MATLAB

Prerequisite

Before starting this section, student should know:

1. A fundamental knowledge on MATLAB
2. A theoretical knowledge on z-transform

Lab Outcome

After Finishing this lab student should be able to:

1. Understand difference equation transformation into z domain, pole-zero representation and system impulse response from transfer function using MATLAB
2. Understand stability of Discrete time system using MATLAB

4.1 z-Transform

The Z-transform converts a discrete time domain signal, which is a sequence of real numbers, into a complex frequency domain representation.

The Z-transform of a discrete-time signal $x(n)$ is the function $X(z)$ defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

where n is an integer and z are, in general, a complex number.

Example 4.1:

Find the z-transform of $x(n) = a^n u(n)$

z-transform using MATLAB code

```
clc; close all; clear all;  
%% z-transform using MATLAB example 4.1  
syms a n x Y  
x = a^n;  
X= ztrans(x)
```

Output of Example 4.1

$X = -z/(a - z)$

[This is the output of command window and ztrans is always used in causal case only.]

Example 4.2:

Find the z-transform of $x(n) = n(0.5)^n u(n)$

z-transform using MATLAB code

```
clc; close all; clear all;
%% z-transform using MATLAB example 4.2
syms n x Y
x = n*(0.5)^n;
X= ztrans(x)
```

Output of Example 4.2

$X = (2*z)/(2*z - 1)^2$

[This is the output of command window and ztrans is always used in causal case only.]

Example 4.3:

Find the inverse z-transform of $\square(x) = \frac{z}{(-0.2)(-0.6)}$

Inverse z-transform using MATLAB code

```
clc; close all; clear all;
%% inverse z-transform using MATLAB example 4.3
syms z X
X=z/((z-0.2)*(z-0.6));
x= iztrans(X)
```

Output of Example 4.3

$x = (5*(3/5)^n)/2 - (5*(1/5)^n)/2$

[This is the output of command window and iztrans is always used in causal case only.]

4.2 Pole Zero Representation and Transfer Function from poles and zeros

The poles are those values for which the system transfer function becomes infinite. These are the roots of the denominator of a transfer function. The zeroes are those values for which the system transfer function becomes zero. These are the roots of the numerator of a transfer function.

$$X(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{a_0 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

Effect of poles on stability.

- (i) **Poles outside the unit circle ($|z| > 1$):** lead to exponential growth even if the input is bounded.
- (ii) **Multiple Poles on the unit circle:** Always results in a polynomial growth.
- (iii) **Simple (non-repeated) Poles on the unit circle:** Can lead to an unbounded response.

The region of convergence (ROC) of $X(z)$ is the set of all values of which $X(x)$ attains a finite value. Thus, any time we cite a z-transform we should also indicate its ROC.

Effect of ROC on stability and causality.

The ROC of stable LTI system must include the unit circle

- (i) **Stable causal system:** All the poles must lie inside the unit circle.
- (ii) **Stable anti-causal system:** All the poles must lie outside the unit circle.
- (iii) **Stability from Impulse Response:** $h(n)$ must be absolutely summable and the summation must be less than infinity.

Example 4.4:

For a transfer function of $\square(z) = \frac{1}{1-0.9z^{-1}}$, give the pole zero representation.

Pole-Zero representation from transfer function using MATLAB code

```
clc; close all; clear all;
```

```
% pole-zero representation using MATLAB example 4.4
```

```
b=[1];  
a=[1 -0.9];  
zplane(b,a)
```

Output of Example 4.4

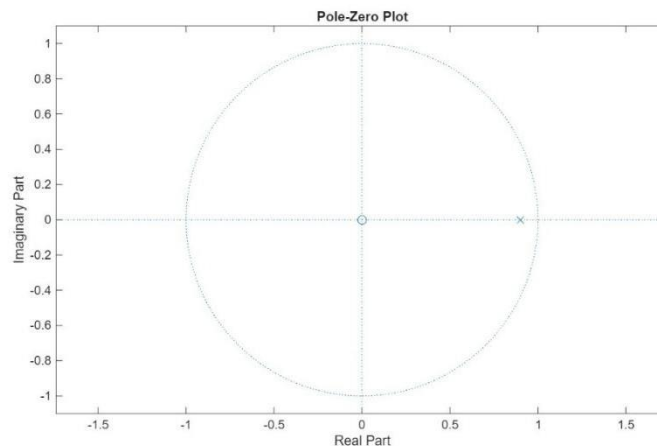


Figure 1: Pole and zero in zplane

Example 4.5:

Find the transfer function from the pole, $p=0.9$, zero, $z=0$ and gain=1.

Transfer function from poles and zero using MATLAB code

```
clc; close all; clear all;  
% transfer function from pole-zero using MATLAB example 4.5  
z=0;  
p=0.9;  
k=1; % gain  
[b,a] = zp2tf(z,p,k)
```

Output of Example 4.5

b = 1 0

a = 1.0000 -0.9000 [This is the output of command window. It shows the input and output coefficients for transfer function.]

4.3 System Analysis Using Pole-Zero

Example 4.6:

For the difference equation, $y(n] = \frac{1}{2}y(n - 1) + x(n]$, of an Accumulator; find the followings:

- Plot the pole-zero in z-plane.
- Find the impulse response $h(n]$.
- Comment on stability.

Pole-zero representation from difference equation using MATLAB code

```
clc; close all; clear all;  
%% Accumulator System Response  
b=[1];  
a=[1 -0.5];  
zplane(b,a)  
title("Pole-Zero in zplane for Accumulator");  
  
h=impz(b,a);  
figure  
stem(h)  
title("Impulse Response for Accumulator");  
S= sum(abs(h))
```

Output of Example 4.6 (i)

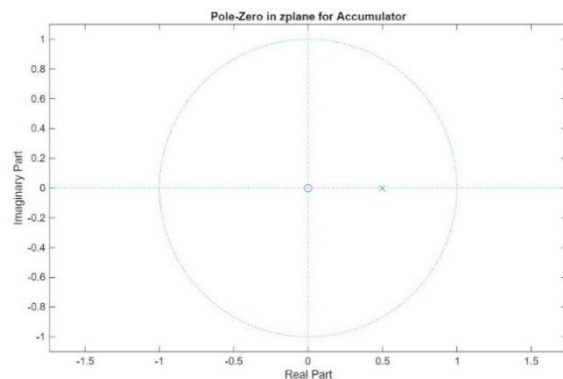


Figure 2: Pole and Zero of an Accumulator in zplane

Output of Example 4.6 (ii)

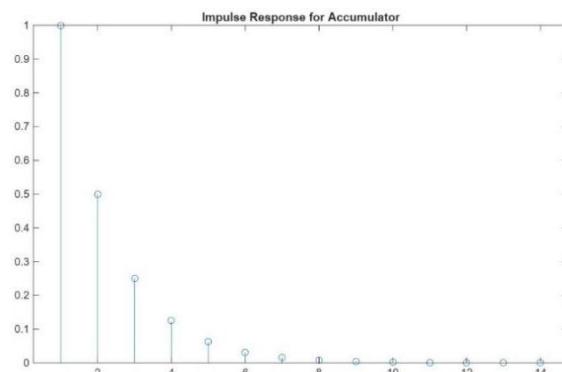


Figure 3: Impulse Response of an Accumulator

Output of Example 4.6 (iii)

S = 1.9999 [This is the output of command window. It shows that the sum of the impulse function $S < \infty$. So, the system is stable.]

Example 4.7:

For the difference equation, $y(n) = \frac{1}{4}[x(n) + x(n-1) + x(n-2) + x(n-3)]$, of a Moving Average Filter (MAF); find the followings:

- Plot the pole-zero in z-plane.
- Find the impulse response $h(n)$.
- Comment on stability.

Pole-zero representation from difference equation using MATLAB code

```
clc; close all; clear all;  
%% Moving Average Filter (MA) System Response  
b=0.25*[1 1 1 1];  
a=[1];  
zplane(b,a)  
title("Pole-Zero in zplane for MA");  
  
h=impz(b,a);  
figure  
stem(h)  
axis([0,5,-0.1,0.3]);  
title("Impulse Response for MA");  
S= sum(abs(h))
```

Output of Example 4.7 (i)

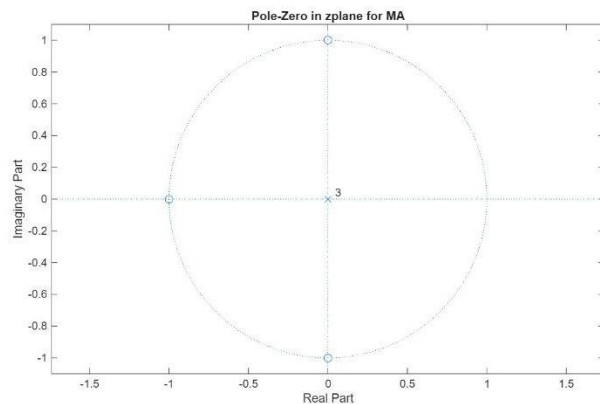


Figure 4: Poles and zeroes of a 4-point Moving Average Filter (MA) in zplane

Output of Example 4.7 (ii)

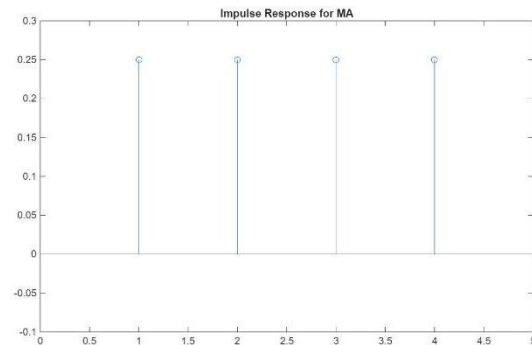


Figure 5: Impulse response of a 4-point Moving Average Filter

Output of Example 4.7 (iii)

$S = 1$ [This is the output of command window. It shows that the sum of the impulse function $S < \infty$. So, the system is stable.]

4.4 System design by analyzing system response from pole and zero

Example 4.8:

Let there be a water-flow sensor. The sensor measures water 2 L/sec. Due to water flow turbulence and air bubble, noise is added to the reference value 2 L/min. Design a filter to reduce the noise while keeping the device at a low cost. Assume, you have 100 sample data from the sensor and the sampling rate to be 1 sample/sec.

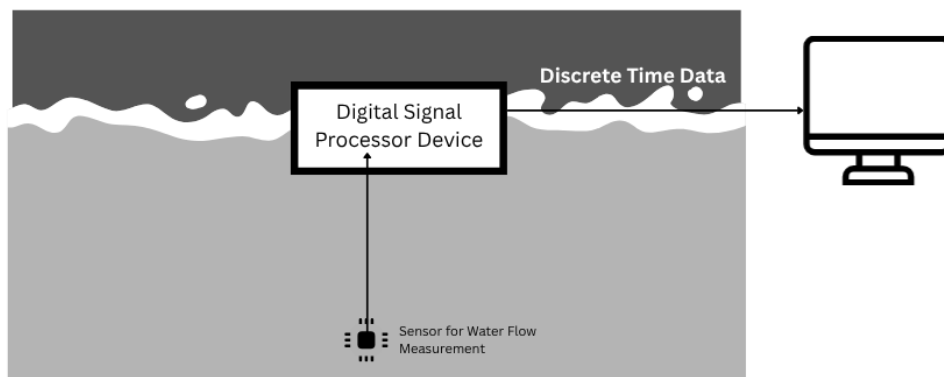


Figure 6: Water-Flow measurement with sensor

- Design a Moving Average Filter and change the filter order “M” to find the most optimized device to reduce noise. Consider M=10 and M=12.
- Design an Accumulator and change the gain “a” to find the most optimized device to reduce noise.
- Compare your results from i) and ii). Choose the best design for the water flow sensing system.

Design of MA filter for water flow sensor noise reduction

```
clc; close all; clear all;
%% Water flow sensor data processing device design
L=100;
x= 2*ones(1,L);
noise= 0.15*randn(1,L);
xn= x+noise;

M= 10;
b=ones(1,M)/M;
a=1;
y= filter(b,a,xn);
error= mean(abs(y(M:end)-x(M:end)))

figure
plot(x)
hold on
stem(xn,'r')
stem(y,'kx','LineWidth',2)
legend('X','Noisy X','Output')
```

Change the value of “M” for different output.

Output of Example 4.8 (i) for M=10

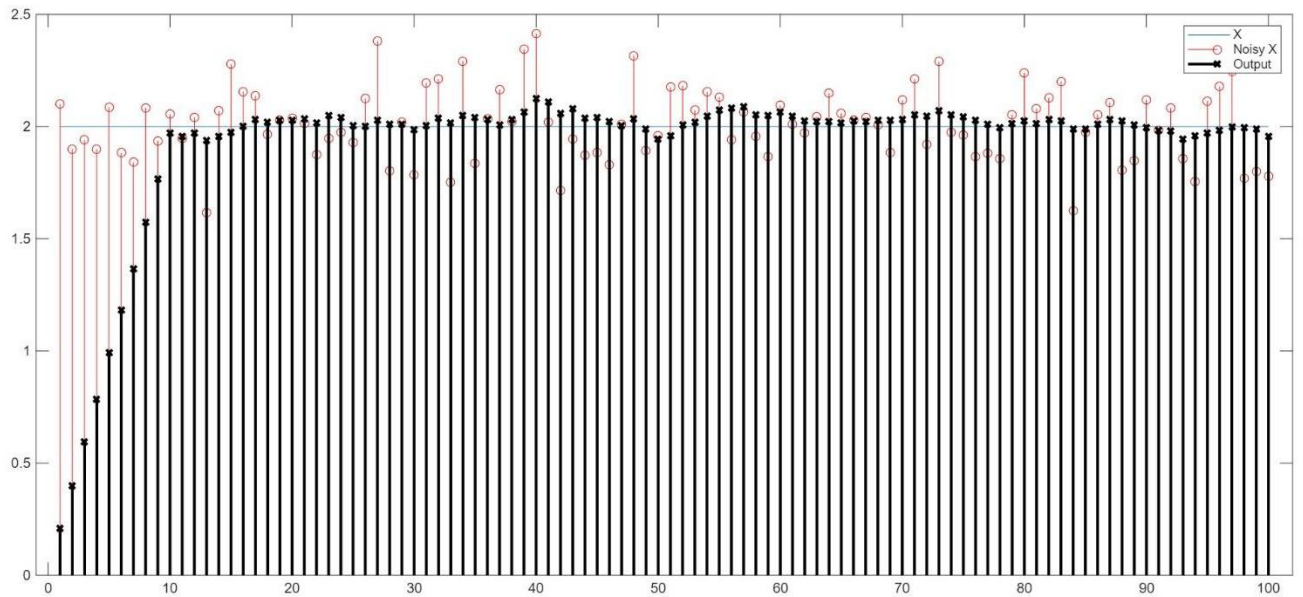


Figure 7: 100 samples of sensor ideal data, noisy data and MA filter processed data with 10-point filter

Output of Example 4.8 (i) for M=12

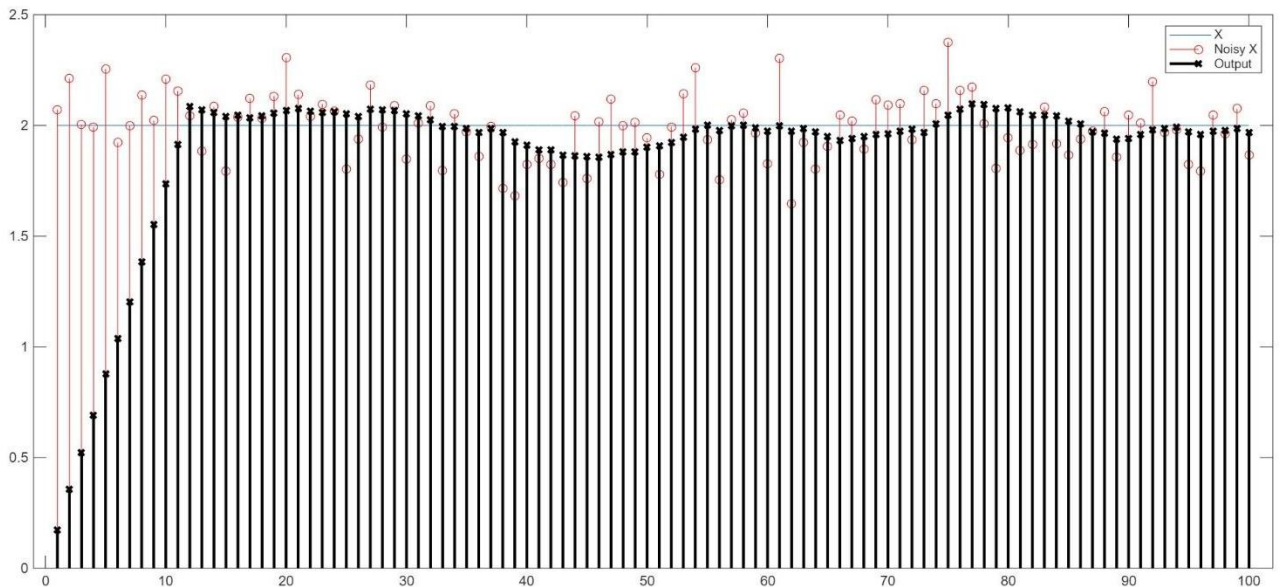


Figure 8: 100 samples of sensor ideal data, noisy data and MA filter processed data with 12-point filter

Output of Example 4.8 (i)

error = 0.0325 [This is the output of the command window for M=10.]

error = 0.0513 [This is the output of the command window for M=12.]

Design of Accumulator for water flow sensor noise reduction

```
clc; close all; clear all;
```

```
% Water flow sensor data processing device design
```

```
b1=1;
```

```
a1=[1 -0.5];
```

```
y1= filter(b1,a1,xn);
```

```
error1= mean(abs(y1(M:end)-x(M:end)))
```

```
figure
```

```
plot(x)
```

```
hold on
```

```
stem(xn,'r')
```

```
stem(y1,'kx','LineWidth',2)
```

```
legend('X','Noisy X','Output')
```

Change the value $a=-0.5$ each time for different output.

Output of Example 4.8 (ii) for $a=0.5$

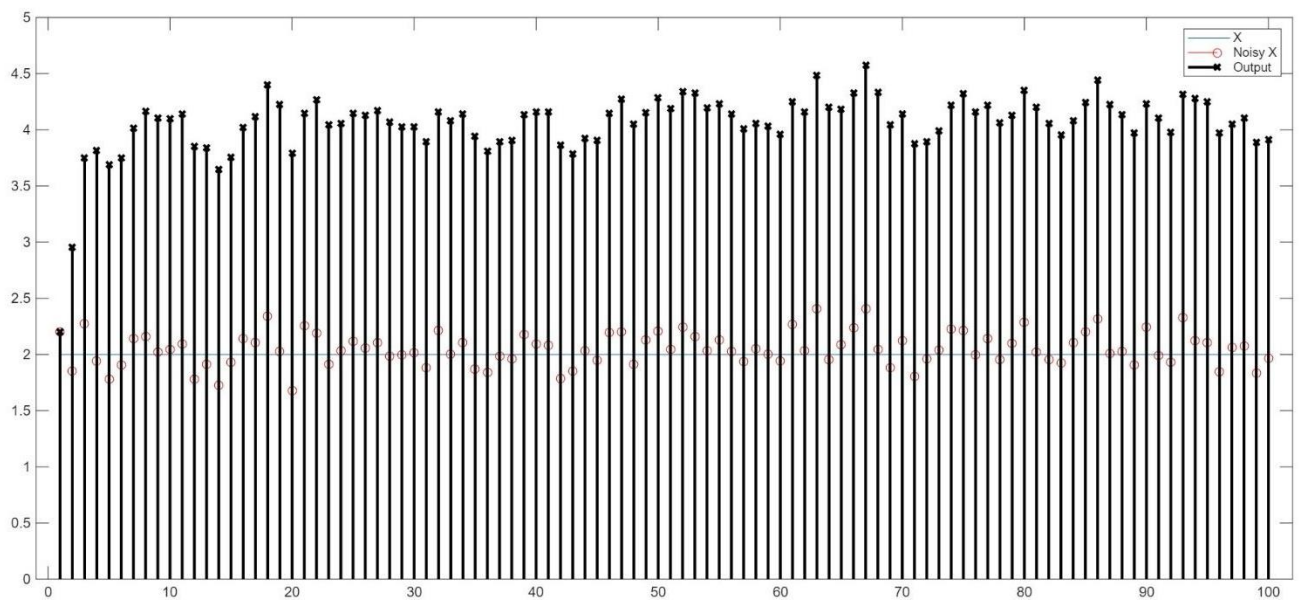


Figure 9: 100 samples of sensor ideal data, noisy data and Accumulator processed data with gain 0.5

Output of Example 4.8 (ii) for $a=0.05$

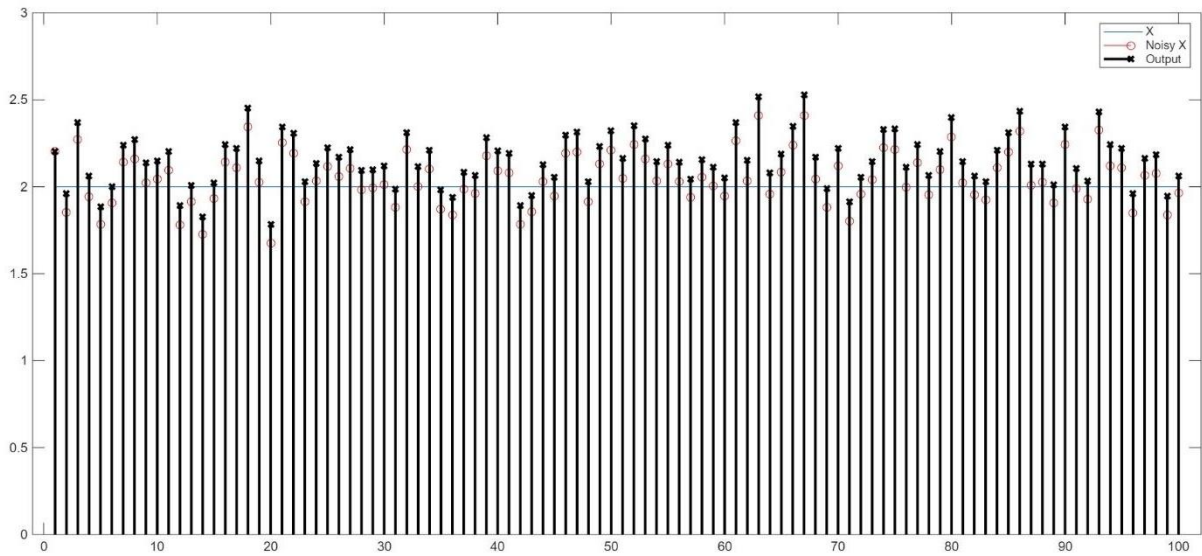


Figure 10: 100 samples of sensor ideal data, noisy data and Accumulator processed data with gain 0.05

Output of Example 4.8 (ii) for $a=0.02$

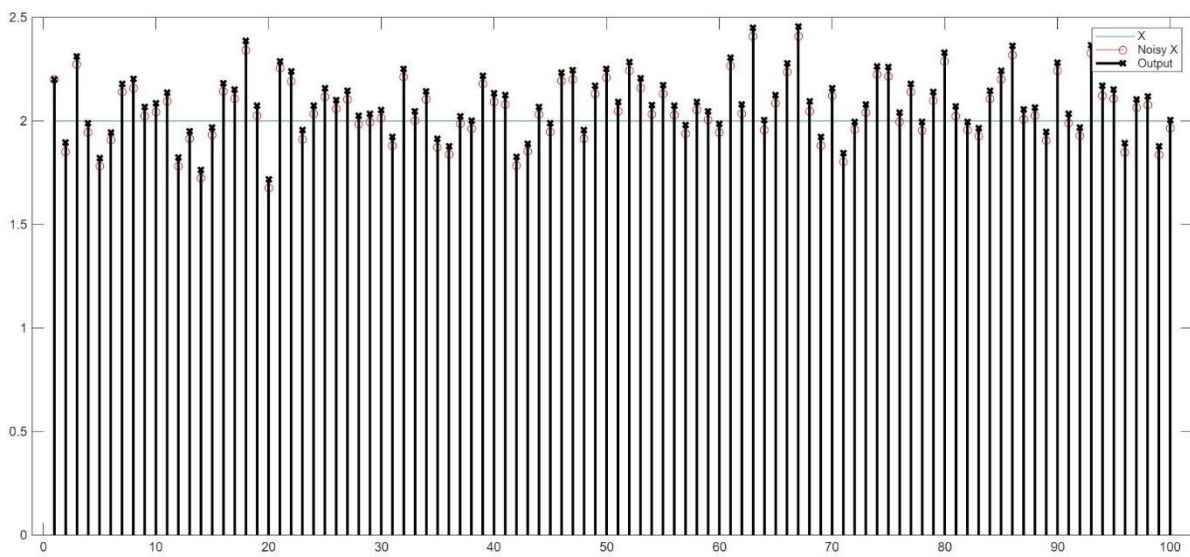


Figure 11: 100 samples of sensor ideal data, noisy data and Accumulator processed data with gain 0.02

Output of Example 4.8 (ii)

error1 = 2.1042 [This is the output of the command window for gain $a=0.5$.]

error1 = 0.1797 [This is the output of the command window for gain $a=0.05$.]

error1 = 0.1383 [This is the output of the command window for gain $a=0.02$.]

Answer of Example 4.8(iii): Even though the MA filter has the least error, the accumulator gives a similar noise reduction effect with fewer components in the system and cost than the MA filter. For this water flow sensor, we may choose the Accumulator.

4.5 Exercise

HW4.01:

Determine the inverse z-transform from the transfer function $H(z) = \frac{1}{1-1.5z^{-1}+0.5z^{-2}}$ using MATLAB command iztrans(). Answer following questions:

- i) Is the system causal?
- ii) What is the ROC of the system?

HW4.02:

Consider the system having difference equation. $y(n) = \frac{5}{6}y(n-1) - \frac{1}{6}y(n-2) + x(n)$.

- i) Find the impulse response $h(n)$.
- ii) Comment on the stability of the system.

HW4.03:

1. Consider the system having transfer function $H(z) = \frac{1-1.2z^{-1}}{1-0.5z^{-1}}$
 - i) Plot the impulse response using MATLAB. Comment on the stability of the system.
 - ii) Plot pole zero on zplane and comment on the stability of the system.
2. Consider the system having transfer function $H(z) = \frac{1+z^{-1}}{1-2z^{-1}+z^{-2}}$
 - i) Plot the impulse response using MATLAB. Comment on the stability of the system.
 - ii) Plot pole zero on zplane and comment on the stability of the system.

HW4.04:

Poles $P_1=1$, $P_2=0.5$ & $P_3=0.2$ and zeroes $Z_1=1$, $Z_2=0.5+j0.886$ & $Z_3=0.5-j0.886$ and the gain $K=1$. Find the transfer function using zp2tf() in MATLAB.